

The Ephemeris

September 2015

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SJAA Tour and Fund Raiser for Lick Observatory

As many of you know, our local Lick Observatory atop Mt. Hamilton, was threatened with termination of funding by the University of California Regents. Despite UC's decision to reverse itself and reinstate it's funding, Lick still faces major financial challenges. SJAA Director, Bill O'Neil, volunteered to organize a private SJAA Tour of Lick Observatory with the purpose of raising awareness and Funds for Lick.

34 generous SJAA members and their guests travelled up the winding Mt. Hamilton Road on May 27, 2015 and paid \$50 per person to raise \$1,700 for Lick. In the last SJAA Ephemeris, there was an article about the recent Google Star Party during which Google announced they were granting Lick Observatory \$500K per year for the next two years, with the hopes of generating matching funds from private donors and other silicon valley companies.

Our host for the evening was Dr. Ellie Gates, one of Lick's resident astronomers, assisted by public program telescope operator Bob Havner. Our Tour included: a presentation in the Lecture Hall about the history of James Lick Observatory (first opened in 1888); a tour of the facility, including seeing the 120-inch Shane reflecting telescope (the main research telescope at Lick) and finally SJAA members got to view astronomical objects through the 36-inch, Great Lick Refractor (this old instrument is a huge 57 feet long!)

Ellie does her research on adaptive optics using a sodium laser attached to the Shane 120", to create an upper-atmosphere target, and a motorized array of mirrors that work to correct atmospheric fluctuations in the final image. The system is being used on the Keck telescopes in Hawaii (also part of the UC Observatories) and is attempting to rival the performance of the Hubble Space Telescope from Earth).

All of our SJAA members who experienced Ellie's tour, were very impressed with her depth of knowledge about the history and research going on at Lick. It was an exciting night and her enthusiasm was infectious. SJAA received many emails on how great the evening was. Here is a small sample of some of the quotes from her fans:

"Ellie was terrific....it was a fantastic evening and a wonderful experience."

"Ellie Gates is one sharp lady. I really thought she delivered a fantastic experience."

"Finally got me up there, and what a thrill. We sure got the cook's tour. Ellie is an excellent teacher!"

SJAA is looking forward to doing a Tour again next year and to finding other ways we can work with Lick in the future. We truly appreciate the research that Lick is doing and want to support their efforts and hope that UC not only continues to support the work at James Lick, but increases the funding to fully staff this wonderful facility. Not only for public outreach, but to facilitate astronomical research, and train future scientists. We need to keep Lick up and running. (see photos on following page)



Sept - Nov 2015 Events

Board & General Meetings

Saturday 9/26, 10/24, 11/21

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Solar Observing (locations differ)

9/19, 10/11, 11/1,

Fix-It Day (2-4pm)

Sunday 9/6, 10/4, 11/1

Houge Park In-Town Star Party

Friday 9/4, 9/18, 10/2, 10/16, 11/6, 11/20

Intro to the Night Sky Class (Houge)

Friday 9/18, 10/16, 11/20

RCDO Starry Nights Star Party

Saturday 9/5, 10/3, 11/7

Binocular Star Gazing

Saturday 10/17

Imaging SIG Mtg

Tuesday 9/15, 10/20, 11/17

Quick START (by appointment)

Friday 10/9, 12/11

Unless noted above, please refer to the SJAA Web page for specific event times & locations.

SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.

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SJAA Tour and Fund Raiser for Lick Observatory



Dr. Ellie Gates

Photo credit: Susan O'Neil



Photo credit: Alex Kramarov



Photo credit: Bill O'Neil



Photo Credit: Les Murayama

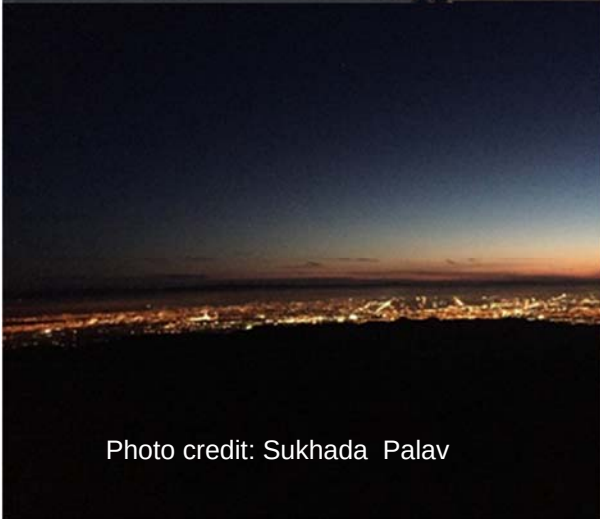


Photo credit: Sukhada Palav



Photo credit: Sukhada Palav



Photo credit: Sukhada Palav

Science

Mars Orbiter - Seasonal Flows in Valles Marineris



Image credit: NASA/JPL-Caltech/Univ. of Arizona

Among the many discoveries by NASA's Mars Reconnaissance Orbiter since the mission was launched on Aug. 12, 2005, are seasonal flows on some steep slopes. These flows have a set of characteristics consistent with shallow seeps of salty water.

This July 21, 2015, image from the orbiter's High Resolution Imaging Science Experiment (HiRISE) camera shows examples of these flows on a slope within Coprates Chasma, which is part of the grandest canyon system on Mars, Valles Marineris. The image covers an area of ground one-third of a mile (536 meters) wide.

These flows are called recurring slope lineae because they fade and disappear during cold seasons and reappear in warm seasons, repeating this pattern every Martian year. The flows seen in this image are on a north-facing slope, so they are active in northern-hemisphere spring. The flows emanate from the relatively bright bedrock and flow onto sandy fans, where they are remarkably straight, following linear channels. Valles Marineris contains more of these flows than everywhere else on Mars combined. At any season, some are active, though on different slope aspects at different seasons.

Future human explorers (and settlers?) will need water to drink, grow food, produce oxygen to breath, and make rocket fuel. Bringing all of that water from Earth would be

extremely expensive, so using water on Mars is essential. Although there is plenty of water ice at high latitudes, surviving the cold winters would be difficult. An equatorial source of water would be preferable, so Valles Marineris may be the best destination. However, the chemistry of this water must be understood before betting any lives on it. Credit: NASA.gov

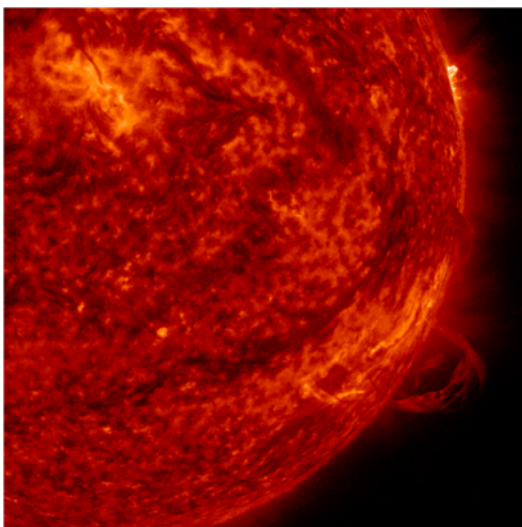
Filaments Break

Over half a day on July 19, 2015, two long threads of solar material called filaments broke free from the sun and erupted into space. The upper one broke first, sending dark plasma down towards an active region below it. An hour or two later, that lower filament erupted. Filaments are huge threads of relatively cool plasma (as compared to the surrounding material) that originate in a low layer of the sun's atmosphere, the chromosphere, and are propped up into the much hotter corona by often unstable magnetic fields.

Once it exploded into space, the second filament eruption grew into an impressive coronal mass ejection, or CME. The Solar and Heliospheric Observatory, or SOHO, which is a joint ESA/NASA mission, observed the CME spreading out into space several hours later.

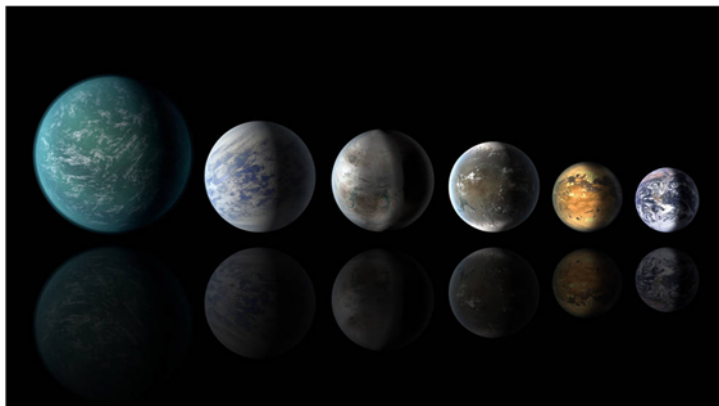
Though CMEs can happen without a filament eruption, a successful filament eruption always precedes a CME. When a filament erupts, it sends a huge cloud of gas, plasma, and magnetic field flying off the sun into space. Sometimes, breaking filaments are not powerful enough to rocket material beyond the sun's gravity, so the gas and plasma fall back to the sun's surface. These events are called failed or partial filament eruptions.

This image is from NASA's Solar Dynamics Observatory, which watches the sun 24 hours a day and can observe extreme ultraviolet light, which is invisible to our eyes. This filament eruption was captured in light of 304 angstroms, typically colorized in red. On the other hand, CMEs are recorded by coronagraphs, which don't look at the sun at all. Instead, they block the bright face of the sun to better observe its faint atmosphere, called the corona.



Science

Finding Yet Another Earth



A newly discovered exoplanet, Kepler-452b, comes the closest of any found so far to matching our Earth-sun system. This artist's conception of a planetary lineup shows habitable-zone planets with similarities to Earth: from left, Kepler-22b, Kepler-69c, the just announced Kepler-452b, Kepler-62f and Kepler-186f. Last in line is Earth itself.

Credits: NASA/Ames/JPL-Caltech

The discovery of a super-Earth-sized planet orbiting a sun-like star brings us closer than ever to finding a twin of our own watery world. But NASA's Kepler space telescope has captured evidence of other potentially habitable planets amid the sea of stars in the Milky Way galaxy.

To take a brief tour of the more prominent contenders, it helps to zero in on the "habitable zone" around their stars. This is the band of congenial temperatures for planetary orbits -- not too close and not too far. Too close and the planet is fried (we're looking at you, Venus). Too far and it's in deep freeze. But settle comfortably into the habitable zone, and your planet could have liquid water on its surface -- just right. Goldilocks has never been more relevant. Scientists have, in fact, taken to calling this water-friendly region the "Goldilocks zone."

The zone can be a wide band or a narrow one, and nearer the star or farther, depending on the star's size and energy output. For small, red-dwarf stars, habitable zone planets might gather close, like marshmallow-roasting campers around the fire. For gigantic, hot stars, the band must retreat to a safer distance.

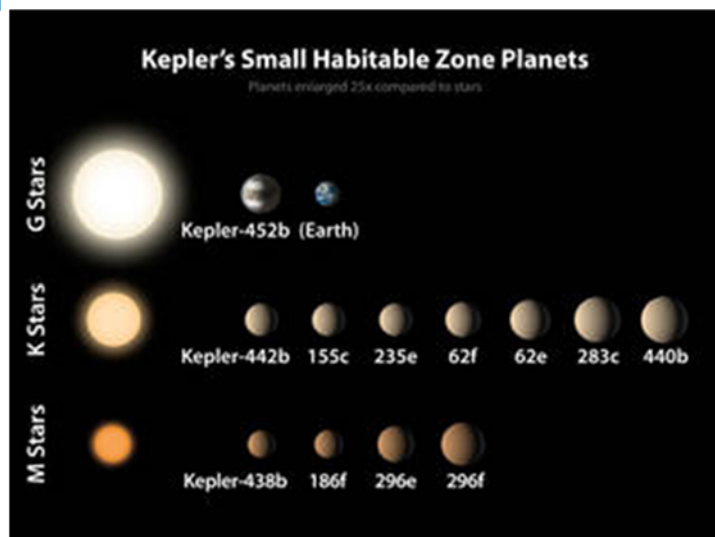
About a dozen habitable zone planets in the Earth-size ballpark have been discovered so far -- that is, 10 to 15 planets between one-half and twice the diameter of Earth, depending on how the habitable

zone is defined and allowing for uncertainties about some of the planetary sizes.

The new discovery, Kepler-452b, fires the planet hunter's imagination because it is the most similar to the Earth-sun system found yet: a planet at the right temperature within the habitable zone, and only about one-and-a-half times the diameter of Earth, circling a star very much like our own sun. The planet also has a good chance of being rocky, like Earth, its discoverers say.

Kepler-452b is more similar to Earth than any system previously discovered. And the timing is especially fitting: 2015 marks the 20th anniversary of the first exoplanet confirmed to be in orbit around a typical star.

Too big, too uncertain, or circling the wrong kind of star: Shuffle through the catalog of habitable zone planets, and the closest we can come to Earth -- at least so far -- appears to be the new kid on the interstellar block, Kepler-452b.



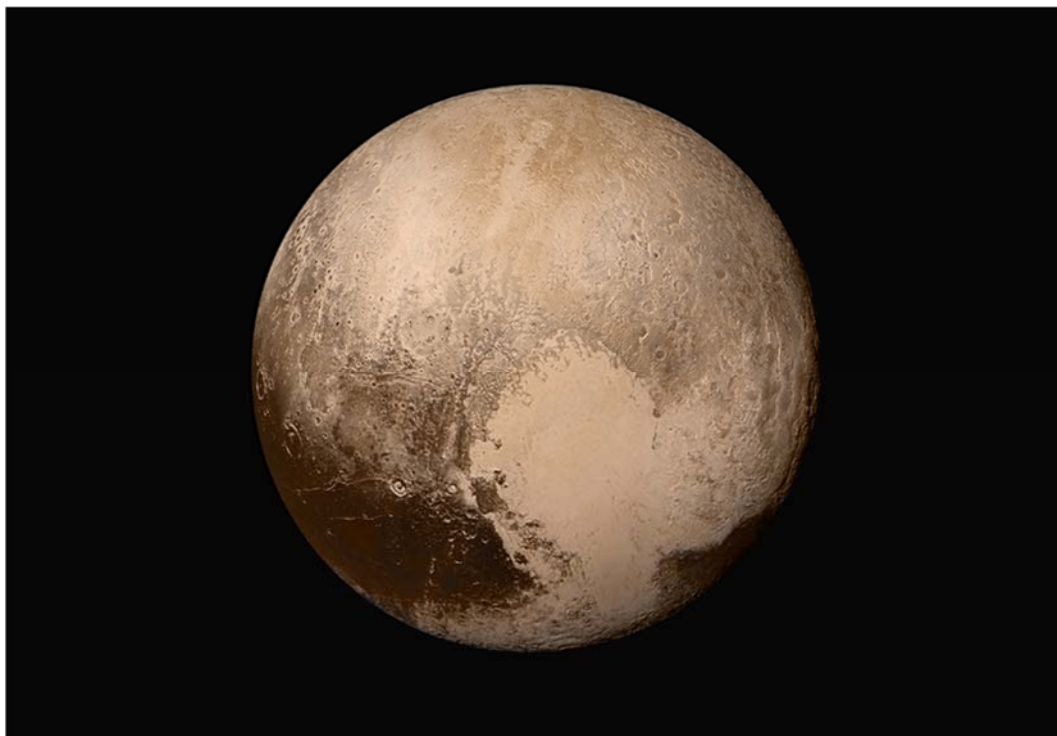
Of the 1,030 confirmed planets from Kepler, a dozen are less than twice the size of Earth and reside in the habitable zone of their host stars. In this diagram, the sizes of the exoplanets are represented by the size of each sphere. These are arranged by size from left to right, and by the type of star they orbit, from the M stars that are significantly cooler and smaller than the sun, to the K stars that are somewhat cooler and smaller than the sun, to the G stars that include the sun. The sizes of the planets are enlarged by 25 times compared to the stars. The Earth is shown for reference.

Credits: NASA/Ames/JPL-Caltech

Credit: NASA.gov

Science

Pluto In True Color



Four images from New Horizons' Long Range Reconnaissance Imager (LORRI) were combined with color data from the Ralph instrument to create this sharper global view of Pluto. (The lower right edge of Pluto in this view currently lacks high-resolution color coverage.) The images, taken when the spacecraft was 280,000 miles (450,000 kilometers) away from Pluto, show features as small as 1.4 miles (2.2 kilometers). That's twice the resolution of the single-image view captured on July 13 and revealed at the approximate time of New Horizons' July 14 closest approach.

Image Credit: NASA/JHUAPL/SwRI

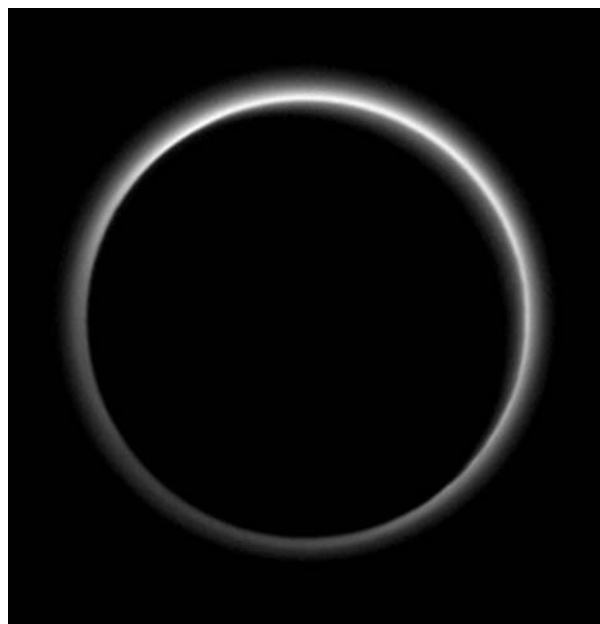
Pluto's Hazy Skies

Speeding away from Pluto just seven hours after its July 14 closest approach, the New Horizons spacecraft looked back and captured this spectacular image of Pluto's atmosphere, backlit by the sun. The image reveals layers of haze that are several times higher than scientists predicted.

Just seven hours after closest approach, New Horizons aimed its Long Range Reconnaissance Imager (LORRI) back at Pluto, capturing sunlight streaming through the atmosphere and revealing hazes as high as 80 miles (130 kilometers) above Pluto's surface. A preliminary analysis of the image shows two distinct layers of haze—one about 50 miles (80 kilometers) above the surface and the other at an altitude of about 30 miles (50 kilometers).

Studying Pluto's atmosphere provides clues as to what's happening below. The hazes detected in this image are a key element in creating the complex hydrocarbon compounds that give Pluto's surface its reddish hue.

Models suggest that the hazes form when ultraviolet sunlight breaks apart methane gas, a simple hydrocarbon known to reside throughout Pluto's atmosphere. The breakdown of methane triggers the buildup of more complex hydrocarbon gases, such as ethylene and acetylene, which were also discovered at Pluto by New Horizons. As these hydrocarbons fall to the lower, colder parts of the atmosphere, they condense as ice particles, forming the hazes. Ultraviolet sunlight chemically converts hazes into tholins, the dark hydrocarbons that color Pluto's surface.



Credits: NASA/JHUAPL/SwRI

SJAA Member Observation Posts

Sixteen Degrees South

From Rob Jaworski

Hello Observers.

I just got back from a holiday in northern Australia. Though it wasn't an astronomical vacation at all (brought no scopes or astronomical meaningful binoculars, I did get a chance to look up at a far more southern sky than I'm used to seeing.

The trip was a family vacation to tropical northern Queensland, specifically, Port Douglas and Cairns, so we were mostly between 16.5 and 17 degrees south latitude, definitely in the tropics, but not all that far south.

The first thing that I noticed was how high Scorpius is at that latitude. It's practically overhead, along with the teapot, with the Milky Way running pretty much between the two. While shielding my view from the bright and numerous resort lights, the sky was still noticeably dark, which allowed the Milky Way to be so visible. What also stood out was a splotch of fuzz between the scorpion tail and the teapot spout, which is known as the open star cluster M7. It was very bright, very noticeable with the naked eye.

After I got over my fascination of how high these summer, er, winter constellations were, I turned my focus to the unknown area just below Scorpius (would that be south or Scorpius, or maybe south west?). It was entirely unfamiliar, and I knew that was a completely new area of the sky that I've never seen before. That was a really cool feeling, knowing that I was looking at something I've never seen before, something I can't see from home because it's over and beyond my familiar horizon.

Checking the southern hemisphere sky map (thanks for sending it, Ed!), I realized that it was mostly Centaurus that occupied that space. This constellation, according to Wikipedia, is visible only up to about 25 degrees north. Its two brightest stars, alpha (yes, *that* Alpha Centauri), and beta, which, btw, are unmistakably bright, point directly to Crux, the southern cross. Crux is recognizable as a cross, and the "fifth" star of the

cross that's depicted in Australia's national flag, though readily apparent in the sky, isn't all that noteworthy. I would opine that it could easily have been left out of their flag and ensigns. I have to go with New Zealand (and a few others) on this one.

The other constellations I identified were the southern crown (Corona Australis) and Triangulum Australe.

After doing a couple of brief (ten or fifteen minute) naked eye observation sessions over a few days, I wanted to see if there was anyone local that I could meet up with to get a better tour, possibly a telescopic one, of the views from sixteen degrees south. Turns out that there is an astronomy group in Cairns, but unfortunately, emails to their listed address went unanswered while I was there.

Of course I wish that I had brought some gear and prepared better for this visit to the south, but I have to remind myself that I went in knowing that this was not going to be an astronomy focused trip. The scuba diving at the Great Barrier Reef, visiting the rainforest as far north as Cape Tribulation, and avoiding the pademelons, wallabies and kangaroos on the road at dusk kept me busy. It's a really interesting area of Australia to visit, where the rainforest meets the reef, as they say, and the lack of astronomy during this visit only gives me an excuse to go there again, perhaps further south, so that the Magellanic Clouds can also be bagged!

Observing during a power outage :)

From Natti Pierce-Thompson

On August 16th, there was a large (3,200 customers) power outage in Gilroy. With the house smelling like vanilla candles, I grabbed my telescope (90mm Mak-Cass) and viewed a bunch of objects from my backyard. Saturn was beautiful, along with Antares the Variable Star in Scorpius. I couldn't believe I could see Messier objects: M4, M13, M92, M22, M28, M7, M17, and even M57 the Ring Nebula!! Woohoo! Oh and one shooting star too. Something fun to do during your next outage!

SJAA Member Observation Posts

Haleakala National Park

From Mark Toney - April 2015

Having the good fortune to have been invited to present the keynote address at the Maui Energy Conference in March, I decided to arrive a couple of days early to squeeze in some viewing on March 22 and 23.

I brought with me a 10 inch dob travel scope purchased a year ago custom-built and hand-ground by Eric Shradler. Aside from the aluminum truss poles that fit into checked luggage, the rest of the scope deconstructed to fit as an airline-compliant carry-on.

Anywhere above 45 degrees, the F5 design provided enough height to comfortably view skies slightly hunched over with my hand on my knees. However, with the scope at low altitudes, especially with viewing southern hemisphere sights close to the horizon, I needed something low to sit on. Fortunately my hotel room provided me with a folding luggage stand that was perfect to sit on, and a cushion from the lounge chair that I could place upon the luggage stand for additional height, or set upon the ground to sit with comfort to view object near the horizon.

The spectacular drive to the summit of Haleakala took about an hour and a half from my hotel room, much of the way crawling up switchback at 15 mile per hour. By the time you hit 8,000 feet in elevation you break through the clouds and by the time you summit at 10,000 feet you are looking down upon the clouds as though from an airplane or the scene from Cloud City in the Empire Strikes Back.

I carried the scope and equipment up the stairs to set up near the observation platform in order to gain an unobstructed view of the southern skies. On Sunday, my first night I shared views with a few tourists until about 8:00pm and was pretty much on my own until I departed at 12:30am. Interestingly enough, on Monday there were more people around, some of which stayed until

almost 1:00am when I packed up to leave. It was particularly fun to hang out with an astronomer from Chile, and another from Brazil, who were thrilled with sights of M81/82 and M51, which they had never before seen.

The elevation and the darkness on the summit of Haleakala rewarded me with the most outstanding views I have ever experienced. Orion's Nebula through an Nagler 9mm exhibited nice bluish hues near the center fading to coppery nebulosity that extended far beyond the wing shapes I usually discern. I was particularly excited to view southern hemisphere objects simply not visible from Northern California. What a thrill to view the Eta Carina Nebula, Jewel Box, and the Southern Cross for the first time.

Nothing, however, prepared me for the breathtaking view of Omega Centauri. When Ed Wong pointed out Omega Centauri to me when we fate brought us together last year in Gualala, it appeared as a faint fuzz ball just barely over the horizon. From Maui, however, Omega Centauri through a 24mm TV Pan was so dense with stars that the core appeared to bulge towards me in a three dimensional effect that made it appear almost like a ping pong ball wrapped in bright white string. Viewing Omega Centauri through the 9mm provided even more stunning views. It appeared as though I could resolve over a thousand individual stars though I cannot profess to having counted them all.

Star party at Glacier Point

From Sukhada Palav

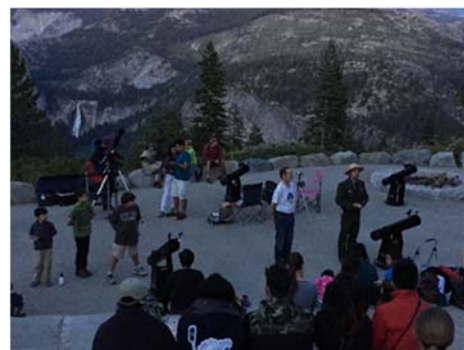
This is no OR as it was completely cloudy! :(But I promised I would report back about my experience up there. So here I go :-)
We made it to Glacier Point well before the sunset for a ranger talk, which was quite interesting. Following the talk though, there were 2 simultaneous star parties planned for the night. The Ranger who gave the sunset talk also delivers a "What's up" session with a

green laser pointer starting at 9pm and people are free to view with their binoculars etc. This happens near the railing. I think this is Yosemite NP's standard star party and I have attended before.

Special for the night was the star party hosted by Sonoma County Astronomical Association up at the Amphitheater. This was the first star party in the summer series at Glacier Point. I think our turn is sometime in August this year! We all took seats in the amphitheater for a brief talk by one of the club members. There was a nice demonstration of solar system by 3 kids (on the left of 1st picture below). These kids won an essay competition held by SCAA and were awarded a telescope each (I think XT4.5) as a prize! I thought that was pretty neat way of encouraging kids...

Overall, there were quite a few telescopes set up but all we could see was the moon. The club spokesperson stretched the talk as much as he could. It was mostly about the Summer Solstice. But eventually, around 9:30pm people started leaving. As it happens all the time, it started clearing up as I drove down to the valley! Sigh... By the time we reached our hotel in El Portal, the sky had completely cleared up and even with bright lights all around me, I could see lot more stars standing in the hotel parking lot than I see from Fremont!

Well, that's all...for now. I am sure I will make it up there again and hopefully, with my telescope!



SJAA Member Observation Posts

New Moon Observing at Mendoza Ranch

From Ed Wong Friday, June 12, 2015

We had about 10 people last night. It was a nice size group. The night started out with the seeing not that great, Sky Quality Meter (SQM) reading of 20.70, it seemed like it took a while to get dark after the sunset. We looked at the usual stuff as it was getting dark. I was using my 4 inch William Optics refractor, Venus was at half phase now. Jupiter and Saturn were up, the views were ok but there was a lot of turbulence so not that sharp.

After looking at the planets, I started in on my plan for the night to look at carbon and binary stars.

- 24 Comae Berenices - nice yellow and white pair, was pretty view at 150x
Y Canum Venaticorum - La Superba - very red carbon star, nice to see it again it's in Canes Venatici
Alpha1 Canum Venaticorum, Alpha2 Canum Venaticorum - another very nice yellow and white pair in Canes Venatici
- Xi Bootes - another very nice yellow and white pair in Bootes
70 Ophiuchi - Nice twin yellow pair of stars in Ophiuchus
- Albireo - A nice view of the well know favorite in Cygnus
- The Double Double - double your pleasure of binary stars with this twin set. Had to put on my 2.3mm eyepiece (300x) in order to get the split on this nice pair.

The conditions had gotten slightly better by this time. I took an SQM reading of 20.82

- Omicron1 Cygni, Omicron2 Cygni - another nice blue and yellow pair like Albireo in Cygnus
- 61 Cygni - Another very nice twin yellow pair of stars in Cygnus
- V Aquilae - Very red carbon star near the M11 the Wild Duck Cluster

As the evening progressed, the viewing conditions got a little better; I took one last SQM reading of 20.93. By this time the Milky Way was visible. So I switched to my 15x70 binoculars and climbed up through the Milky Way viewing along the way:

Collinder 316>

M7>M6>M8>M22>M20>M21>M23>M24>M25>M17>M16 and finishing at M11.

If you have not tried this with a pair of binoculars it's well worth the effort. 13 very nice objects just in that one section of the sky.

As we started packing up around midnight, we were looking at the night sky and comments were made that it, seemed

like the sky glow to the southwest and west seemed worse than other nights. One possible reason could be that recently Gilroy has installed a number of LED street lights since the Club first started observing at Mendoza Ranch. Regardless though it was a fun night.

Also, I need to mention and big congratulations to Ryan Lu who came last night. He told me he had been working on Astronomy League Herschel 400 list and was able to complete it last night. Great Job Ryan!!

Ok that's it for now, until next time under the stars.

Fremont Peak

From Mark McCarthy June 12, 2015

Friday night I went up to the Peak to catch-up on my H400 list. Having spent most of my March and April sessions on Herschel Sprints, and being clouded out in May, I was surprised to notice at how fast the sky had moved. I had many targets in UMa, Cvn, and Vir, but they are all setting soon after astronomical twilight. I used to never pay attention to the changing of the constellations, but now it's of primary concern in my astronomical life planning.

This part of the list is dominated by elliptical galaxies, so many of which have the same description (stellar nucleus, glowing core), differing in their brightness, degree of elongation, and position angle. I try not to get bored with them, keeping in mind their vast distance, the wonder of light, and thinking of all the life which could be lurking in there. All the same, it was a delight to come across striking and unusual objects on the trail, along with quite a few लग्नappes. All with 20" f5.25 reflector, 121x, 0.7°.

Transparency was excellent but seeing was average; it was windy all night and at times my scope weathervaned while I was taking notes or checking my chart.

- NGC 3998, 3982, plus 3990 & 3972, UMa: Four galaxies in FOV!.....
- NGC 4026, UMa: Bright round ball of a core with a long thin halo
- NGC 4631 plus 4627, Cvn: Wow! Huge, long and thin.....
- NGC 4656 plus 4631, Cvn: Immediately saw as two galaxies, both with dim halos with flowing cores.....
- NGC 4485 & 4490, Cvn: Wow! 4490 is larger, elliptical SSW-NNE, bright core and mottled halo....
- NGC 4449, Cvn: A nice ragged, mottled mess!
- NGC 4085 & 4088, UMa: 4085 is large...Nice!
- NGC 4565, Com: Wow! Beautiful long edge on...

For the ride home in the morning, I had the company of a cricket which had stowed away in my gear. I released it in my backyard upon arriving home, hopefully to keep me and my family company through the summer.

Perseids Meteor Shower at Coyote Valley

From Vini Carter



Coyote Valley Open Space Preserve

Once a year the Bay Area plays host to one of the very best meteor showers. It's called the Perseids because the meteors appear to emerge from the constellation Perseus and they streak across the sky. This year, we have a brand new preserve to host it in!

Editor's note: this announcement is from the Open Space Authority (OSA) webpage <http://www.openspaceauthority.org>. The SJAA was asked by OSA to volunteer telescopes for the event. At the far end is Lee Hoglan with Armstrong's brass Renaissance APO. Natti Pierce Thompson is near the middle with her small red Maksutov. Vini and Leslie Carter are on the near end (left) with their Sky-90 next to Gary Chock.

Photo credit: Vini Carter

By the time things got into full swing, the OSA calculated between 800 and 1000 people came. They probably had five to seven volunteers just directing automobile traffic. You can see the scopes and their owners in the photos (attached pdf). There are more photos on the meet-up site from other photographers.

It was an amazing night; for me, much better than Pinnacles. (Unfortunately the public kept dragging me back to Saturn, which was pretty good, but...) The Milky Way was so bright that it was obvious to every one in attendance. Lots of open sky, and room to move (no crowding around the scopes). It was fun to hear a whole field of people going "ooooo" and "aaaahh" with each shooting star.

Never saw a cloud while at Coyote Valley, but when we got back to HP (at 1:00 am) the sky was completely overcast. Nonetheless, as we unloaded the car, I saw one shooting star so bright it was clearly visible through the overcast.

These last few evenings have been wonderfully clear in the San Jose area; great for backyard viewing. Think I'll head out to the back yard right now!

We had direct line-of-sight to Lick Observatory(s) on Mt. Hamilton. I used one of the domes to check the calibration of my finder scope!

Imagers (without scopes) put their cameras and tracking mounts further out in the field, to get a silhouette of the hillside against the milky way.

There are some beautiful astro images posted on the OSA Meet-up site as well <http://www.meetup.com/OSA-Hiking-Enthusiasts/events/223772802/>

SJAA Pinnacles Outing

From Ed Wong

The SJAA had it's first overnight camping star party at Pinnacles National Park on July 10th & 11th. Ed Wong organized the two-night campout/star party. It was open to SJAA members and immediate family members. We ended up with five group campsites at Pinnacles, thanks to Glenn Newell and Dwight Shackelford for booking their own group campsite which allowed us to have more parking and room to set up. We had forty people at the event to enjoy the night sky at one of the darker sites that's still relatively close to the bay area.

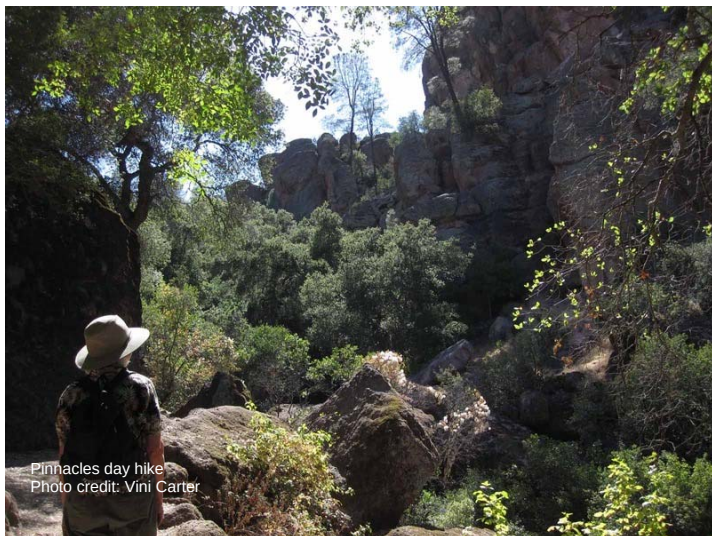
With 5 campsites, there was enough room for us to setup our tents and gear with room to spare. It was nice for me to see peo-



Photo credit: Tom Piller

ple I have not seen in a long time and meet new members of the club I have never met. For me the social aspect of the event was one of the highlights. Fortunately for us the weekend was not as hot as it normally is. Temperature was in the high 80's during the daytime. Most of that were at the campgrounds during the daytime gathered under the big tree in the camp grounds and told stories as astronomers will do during the daytime :-). Others ventured off to do day hikes and explore around the Pinnacles.

One of the highlights of the trip from a social aspect is the meal times. Kudos to Teruo Utsumi for organizing the meals. Teruo with the help of others prepared some great meals which included



Pinnacles day hike
Photo credit: Vini Carter

steak and ice cream one night for dinner (yum yum) it was nice to gather around to socialize and the sun was going down.



L>R Leslie & Vini Carter, Teruo Utsumi and (unknown).....dinner time
Photo credit: Tom Piller

The first night of viewing started out a little hazy, there was a thin layer of clouds for about the first hour as the atmosphere was stabilizing. I took an SQM reading of 21.40. As the night cleared up the viewing got better. I took another SQM reading of 21.65 later in the evening. The Milky Way is much more visible at this site; one could see it span all the way through Cygnus.

I viewed mostly with the binoculars and began working my way down the list that I had prepared for the group for the Deep Sky Scavenger Hunt.

I first observed the Markarian Chain then Mel 111, M64, IC4665, IC4756, the Coat Hanger, Veil Nebula, North American Nebula, Andromeda Galaxy and M33 but I ended up socializing most of the night, helping other people find objects. It was a fun night for me.

The second night was clear from the beginning, once it got dark. I got started looking for the binary stars in the DSSH list with my telescope. There were a lot more of the public this night walking around with flashlights so it made it harder to get dark adapted but I was able find a fair amount of the binary stars I was looking for. This was also a good night to socialize. The SQM reading got up to 21.68.

The Pinnacles trip was fun. The feedback from most people who attended was good.

The things that people wished for most were less interruptions from the public with the flashlights and more viewing time. I'm working on trying to make those two wishes come true.

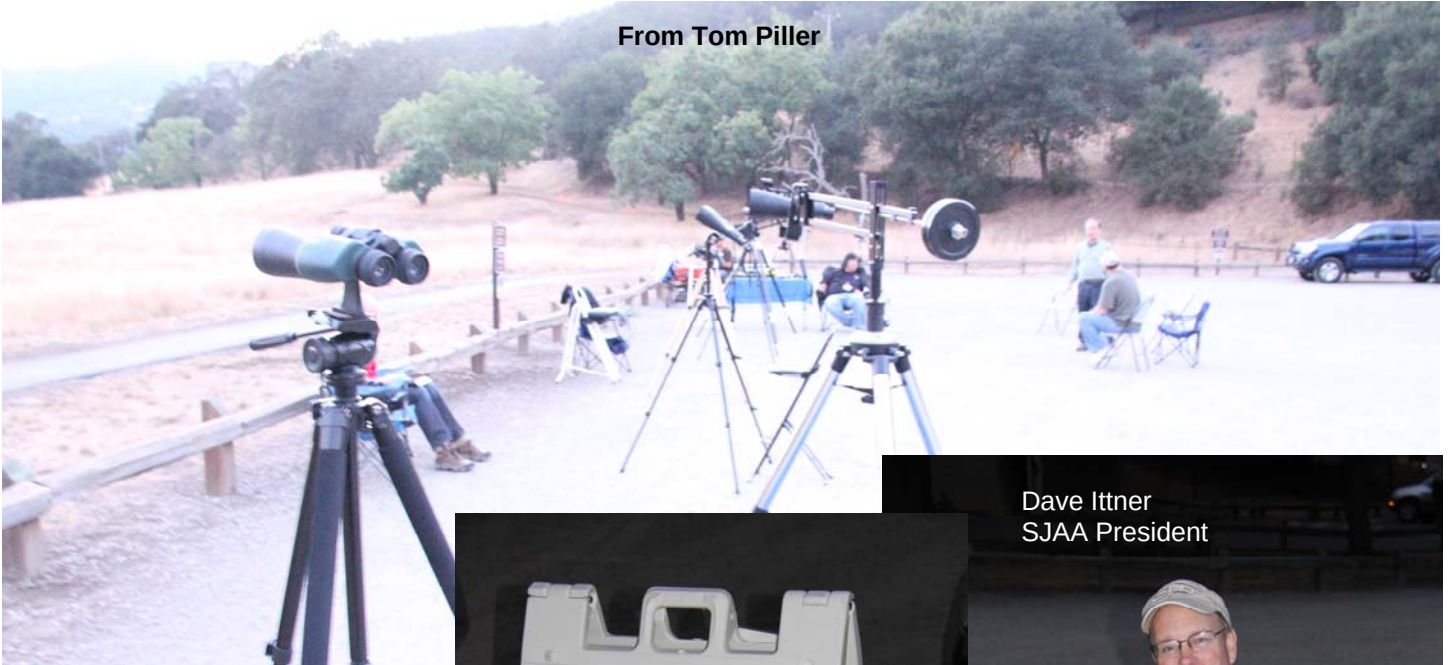
We may have to try again in the late fall to get more viewing time since the sun will set earlier then. Overall though, for a two hour easy drive from San Jose to get to some darker skies it was worth it. I know for some that went, that was the first time they have viewed under dark skies.



Binocular Star Gazing at RCDO

8-22-2015

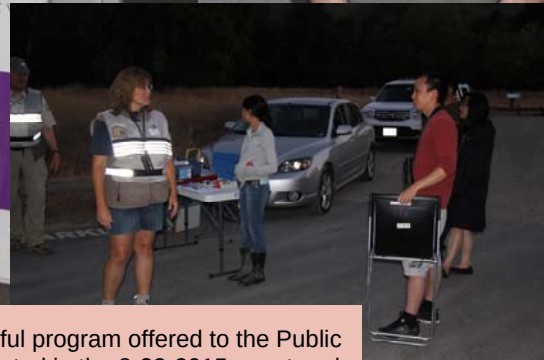
From Tom Piller



Dave Ittner
SJAA President



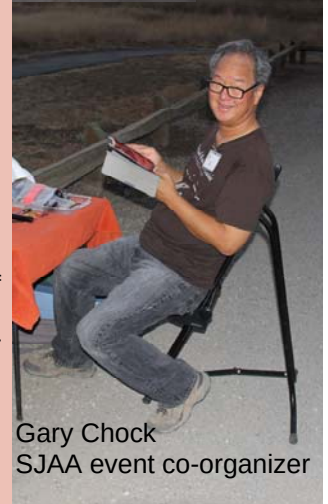
Ed Wong
SJAA Director and event organizer



Binocular Star Gazing at Rancho Canada Del Oro (RCDO) is a successful program offered to the Public by SJAA in conjunction with the Open Space Authority (OSA). I participated in the 8-22-2015 event and was impressed with how well the event was received by the enthusiastic guests. Most were newbies and all very interested in learning about night sky observing.

Created by Ed Wong, this program's mission is to demonstrate to interested budding astronomers that: one doesn't have to spend a bundle on equipment to be able to view the night sky; a pair of \$20 binoculars can get you to the party. learning some of the first steps in successful night sky observing like using printed maps of the night sky, finding the constellations you want to observe, finding the objects in the constellations, and star hopping to the objects can be fun and rewarding. Learning the most common asterisms in the night sky is a good tool to help zero in on the location of the constellations. Learning that the night sky is constantly changing; it's moving all the time and there are new constellations and objects to observe every season.

I enjoyed watching SJAA teachers Ed Wong, Dave Ittner and Gary Chock guide the guests through a small list of objects, first finding them on paper then aiding them in making that important and sometimes difficult connection from paper maps to the night sky.



Gary Chock
SJAA event co-organizer

SJAA Member Astrophoto Gallery



Photo Above

From: Hitesh Dholakia,
Shashank & Shishir

Subject: Flaming Star
Nebula (IC405 in Auri-
ga)

Location: Lake San
Antonio, CA

OTA: Astro-tech
111mm, F7

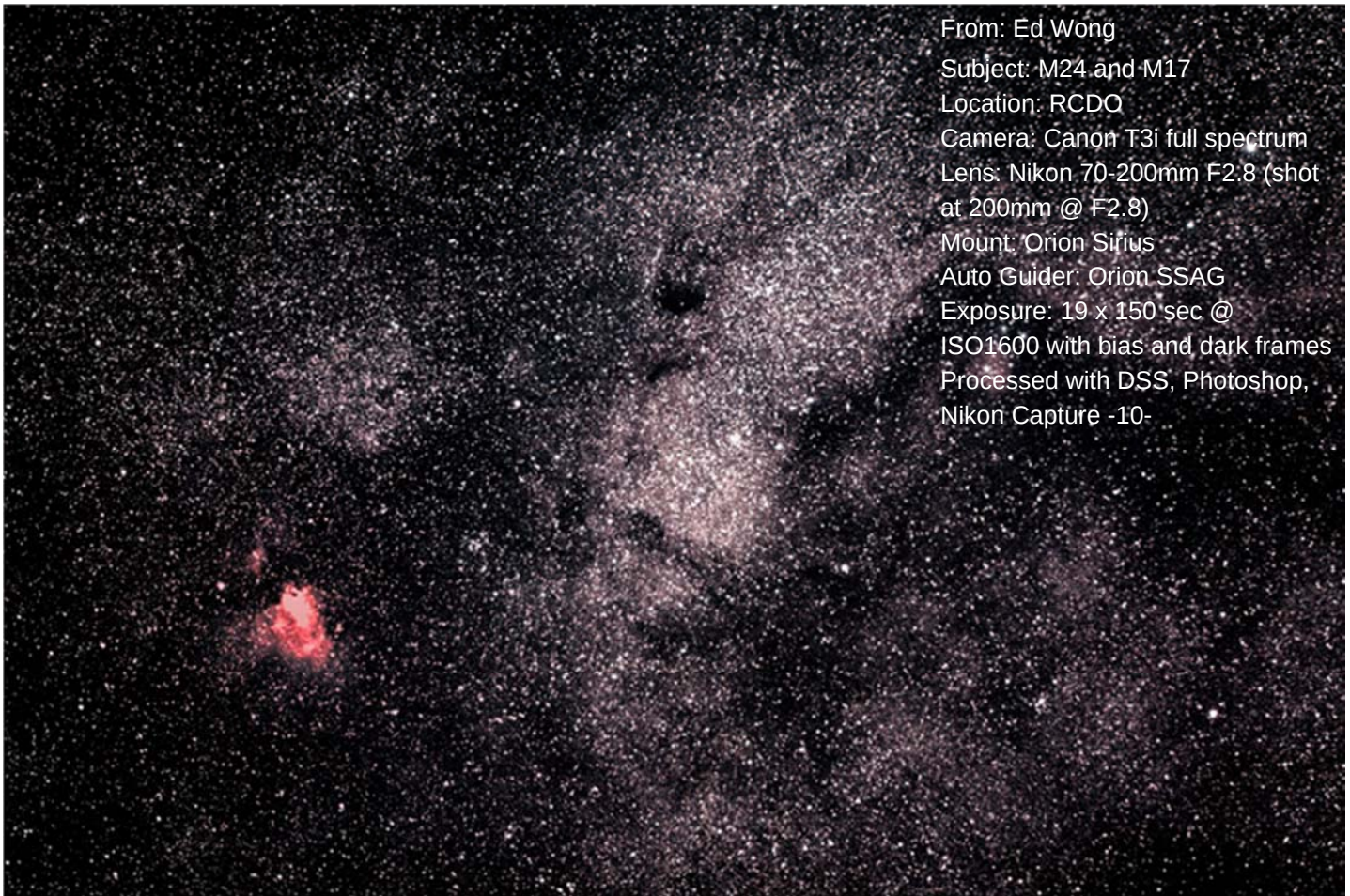
Camera: SBIG ST-
8300M

Mount: Atlas EQ-G

Processed with Deep
Sky Stacker, Pixinsight

From: Marilyn Perry
Subject: July 2015 "blue moon"
Location: Sierra Vista
Date: July 31, 2015
Camera: Canon T3i, 200mm lens





From: Ed Wong
 Subject: M24 and M17
 Location: RCDO
 Camera: Canon T3i full spectrum
 Lens: Nikon 70-200mm F2.8 (shot
 at 200mm @ F2.8)
 Mount: Orion Sirius
 Auto Guider: Orion SSAG
 Exposure: 19 x 150 sec @
 ISO1600 with bias and dark frames
 Processed with DSS, Photoshop,
 Nikon Capture -10-



From: Hitesh Dholakia, Shashank & Shishir
 Subject: Star field near Antares
 Constellation: Scorpius, Ophiuchus
 Location: Likely, CA
 Camera: Canon 60d, 80mm lens
 Exposure: 5x240 second (20 min total), 800 ISO
 Mount: Atlas EQ-G
 Processed with Deep Sky Stacker, Pixinsight

The blue regions are reflection nebulae. The yellow
 regions are emission nebulae. You can also spot the
 globular cluster M4 right of the bright star Antares.

Brahmandam



From: Paul Mahany

Subject: Bubble nebula shot at Manzanita Observatory Yosemite August 8, 2015

Camera: Canon 60Da with Paracorr II

Lens: ATM 6" F5 Newtonian, 400mm

Mount: Atlas EQ6 Atlas

Auto Guider: Orion 50mm

Exposure: 9 hours

Processed with StarTools, DSS, BYEOS, SkyTools

KID SPOT



Kid Spot Jokes:

- ♦ **What kind of light goes around the Earth ?**
A satel-lite
- ♦ **How does the solar system hold up its pants ?**
With an asteroid belt.

Kid Spot Quiz:

A. Meteor showers result from dust trails left by
a) Moon b) Asteroids c) Comets

B. During a meteor shower meteors appear to move outward from a point in the sky. What is this point called?

a) Zenith b) Radiant c) Origin



Image: Hubble space telescope
Photo Credit: NASA

Did You Know?

Source: Astronomy Magazine March 2015 issue
"500 Coolest Things"

- ♦ The most luminous star visible to the naked eye, 34 Cygni, out-shines the sun by 610,000 times
- ♦ If Earth were the size of a tennis ball, the Sun would be a sphere 24 feet across approximately 0.5 miles away.
- ♦ The terms "Tropic of Cancer" and "Tropic of Capricorn" originated several thousand years ago when the Sun stood before the stars of those constellations on the summer and winter solstices, respectively.
- ♦ In 1972, Cygnus X-1 became the first X-ray source widely recognized to be a black hole.
- ♦ By total energy, sunlight is some 50 percent infrared light, 40 percent visible light, and 10 percent ultraviolet light.
- ♦ The five most common elements in the Sun by mass are hydrogen (73.90 percent), helium (24.69 percent), oxygen (0.63 percent), carbon (0.22 percent) and neon (0.17 percent).

Kid Spot Quiz Answers:

A. Comets

B. Radiant

Club Updates

From the Board of Directors Announcements

2015 Annual SJAA swap meet will be held on Sunday, November 22nd this year.

2015 Cal Star will be held Sept 10th thru 13th.

Effective June 27th thru 2017, the open Board seat has been filled by Glenn Newell.

The Houge Park club house now has DSL and Wi-Fi is available.

RASC 2016 Observer's Guide / Calendar

The SJAA will be taking pre-orders for the 2016 RASC "Observer's Guide" and "Calendar" through Oct. 24. We expect the order to arrive in time for the Nov. 21 General Meeting. Stay tuned to the announcement email list and the club website for details.

Board Meeting Excerpts

May 30, 2015

In attendance

Dave Ittner, Rob Jaworski, Ed Wong, Lee Hoglan, Bill O'Neil, Michael Packer, Teruo Utsumi

Absent: (excused) Greg Claytor

Guests: Vini Carter, Manoj Koushik

Non-Profit status

Per Rob the club needs to ensure compliance with state and federal requirements for non-profit status. Rob and Michael are reviewing Robert Armstrong's notes and will take appropriate action.

Dr. Robert Armstrong Scholarship

Per Dave Ittner Linda Shore of the Astronomy Society of the Pacific (ASP) informally agreed to a teacher's scholarship in Robert's name. As a memorial to Robert and in recognition of the donation from his estate, SJAA would fund two \$300 scholarships a year for teachers to take Project Astro curriculum (for an undetermined period) and specify the criteria for the award of the scholarships. ASP would administrate the scholarship. Previously raised ideas (name a scope/mount and/or library after Robert) were also discussed. The board will discuss again at the next board meeting.

Adobe Form Central Replacement

Per Dave Ittner the AFC service we're using for the new member signup web form (including PayPal) will become unavailable at the end of June. Swami is working on a replacement, projected to be complete in a week.

SJAA Coders Source Code License

Per Teruo for the on-going automation effort, we need to choose and authorize a license to use for all developed code. The MIT and Apache licenses allow commercialization whereas GPL prohibits such commercialization. The Coders group recommends GPLv3 and the Board approved.

June 27, 2015

In attendance

Dave Ittner, Rob Jaworski, Ed Wong, Lee Hoglan, Michael Packer, Bill O'Neil, Teruo Utsumi

Absent: (unexcused) Greg Claytor

Guests: Vini Carter, Glenn Newell

Appointment of Glenn Newell to BoD

Glenn Newell to fill open board seat with term of 2017. Welcome aboard, Glenn!

SJAA Clearance Sale

Dave proposed clearance sale on previously donated equipment on Sunday August 16th.

Dave makes motion to approve the clearance sale for August 16, 2015. Michael seconds, approved by all.

SJAA Robert Armstrong Scholarship

Discussion continued on how to honor Robert Armstrong and the use of his monetary donation. Additional new ideas will be discussed during the next month on the BoD portion of the SJAA forum.

Announcement of actions taken by President

- S&T Magazine subscriptions will no longer be offered by the club
- The Jr. Member classification has been dropped
- Google for Non Profits - joining program
- The Battleship Binoculars have been sold
- The Adobe Forms Central service was allowed to expire and has been replaced by custom code.

Non-Profit status

No status update, work in progress.

District 9 Celebrate Cambrian Festival

Rob made a request for support for this event on Sunday August 23rd, 10 am to 4 pm. Rob to follow up.

July 25, 2015

In attendance

Dave Ittner, Rob Jaworski, Ed Wong, Lee Hoglan, Greg Claytor, Bill O'Neil, Glenn Newell, Teruo Utsumi

Absent: (unexcused) Michael Packer,

Guests: Vini Carter, Ken Presti

SJAA Robert Armstrong Scholarship

Further discussion pending. To be continued at August board meeting.

District 9 Celebrate Cambrian Festival

Rob to coordinate SJAA booth for the Cambrian Festival for Sunday August 23 at the Camden Community Center.

Coyote Valley Family Harvest & Feast support

The Open Space Authority is holding their Family Harvest & Feast event Sat. Sept. 19. The SJAA has been asked to hold a solar observing event at Coyote Valley Park.

Advanced Imaging Conference @ SCCC, free table

The Advanced Imaging Conference, to be held at the Santa Clara Convention Center offered the SJAA a free table for Sat. Oct. 17. There is some potential to reach local imagers. To be discussed further.

Member Survey

Ken presented his proposed survey to solicit opinion of past and present members on a variety of topics concerning the club. Ken will update the survey based on feedback from discussion.

Bank Draft Resolution

Dave, Rob and Teruo met with Keypoint Credit Union to open new checking and savings accounts. Rob is working on a draft application.

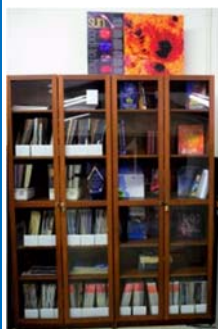
Pinnacles Debrief

The Pinnacles trip earlier this month was very well received. There were many positive comments, both informally and through a post-event survey. We'll consider doing it again next year.

Partnerships with Pinnacles

There are on-going discussions on a potential partnership with Pinnacles National Park and access to their facilities to hold observing events. There will be further updates as discussions continue.

SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astron-

mers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Big thanks to go Vini Carter who has stepped up to take over as Fix It program chair from Ed Wong.

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2:00-4:00 PM at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

<http://www.meetup.com/SJ-Astronomy/>

The September 19th solar observing event will take place at the OSA Coyote Valley Harvest Feast in lieu of Houge Park.

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the

surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are held every other month. The next one is scheduled for October. In addition to the indoor sessions, Ken Presti has offered to hold an outdoor session for the QS attendees to help them learn how to use the loaner scopes and the night sky.

The first field session was held on Saturday, April 18th. There were 13 scopes out at RCDO. Everyone enjoyed the session.

Since it makes no sense to have a Quick Start scope sitting in the storage closet at Houge Park, it is my intention to offer these scopes out to our members with preference to past QS attendees.

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects. We'll need extra help to pull this off, but it appears to be doable."

The announcement and background can be found here:

<https://groups.google.com/forum/#!topic/sjaa-announcements/ScBqsdpmzq>

Loaner Program

The purpose of this program, headed up by Manoj Koushik, is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been

through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



SJAA Contacts

President:	Dave Ittner
Vice President:	Michael Packer
Treasurer:	Rob Jaworski
Secretary:	Teruo Utsumi
Director:	Greg Claytor
Director:	Lee Hoglan
Director:	Ed Wong
Director:	Bill O'Neil
Director:	Glenn Newell
Memberships:	Ed Wong
Beginner Class:	vacant
Fix-it Program:	Vini Carter
Imaging SIG:	Mark Striebeck
Library:	Sukhada Palav
Loaner Program:	Manoj Koushik
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick STARt	Dave Ittner
Solar:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Teruo Utsumi
E-mails:	

<http://www.sjaa.net/contact>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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here

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San Jose, CA 95159-8243

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
- ☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____